

Mapping circuit to transfer function

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August 5, 2026

1 Questions Answered

1.1 solving RC circuit : rc1

so we have a RC circuit, we find its transfer function

$$\begin{aligned}H(s) &= \frac{V_{out}(s)}{V_{in}(s)} \\V_{out}(s) &= V_{in}(s) * \frac{Z_c}{Z_c + Z_R} \\V_{out}(s) &= V_{in}(s) * \frac{\frac{1}{sC}}{\frac{1}{sC} + R} \\H(s) &= \frac{\frac{1}{sC}}{\frac{1}{sC} + R}\end{aligned}$$

$V_{in}(s) = 1$ because input is a AC source with amplitude of 1
say

$$C = 1 \text{ pF} = 1 \times 10^{-12} \text{ F}, R = 1 \text{ k}\Omega = 1 \times 10^3 \Omega$$

for a purely resistive load, we have the following relationship for power vs voltage

$$\begin{aligned}P_{\text{load}} &= V_{\text{out,rms}} \left(\frac{V_{\text{out,rms}}}{R_{\text{load}}} \right) \\&= \frac{V_{\text{out,rms}}^2}{R_{\text{load}}}.\end{aligned}$$

at -3dB cut off frequency, power delivered to a load with resistance R drops

to half.

$$\begin{aligned}\frac{P_{\text{current}}}{P_{\text{max}}} &= \frac{1}{2} \\ \left(\frac{V_{\text{current}}}{V_{\text{max}}}\right)^2 &= \frac{1}{2} \\ \frac{V_{\text{current}}}{V_{\text{max}}} &= \sqrt{\frac{1}{2}} \\ &= \frac{1}{\sqrt{2}} \\ &\approx 0.7071\end{aligned}$$



2 Questions To Be Answered

1. how do they tune VDD for ring oscillator in a real application?
2. poles and zeros in oscillators
3. current mirrors in differential ring oscillator

3 Some Picture